

PRAGATI BHARTI COMPREHENSIVE PHYSICS EXAM

1. What is the SI unit of electric potential difference?

- (A) Ampere (B) Volt (C) Ohm (D) Joule

2. An object is placed at $2F$ in front of a convex lens. The image formed is:

- (A) Virtual and enlarged (B) Real and inverted at $2F$ (C) Real and diminished (D) At infinity

3. In an experiment to observe electromagnetic induction, a solenoid having 500 turns is connected to a sensitive galvanometer and a permanent bar magnet is moved along its axis with constant speed and

the deflection shown by the galvanometer pointer is observed for different speeds.

What conclusion can be drawn regarding the magnitude of the induced EMF?

- (A) Induced EMF is directly proportional to the rate of change of magnetic flux
- (B) Induced EMF is inversely proportional to the number of turns
- (C) Induced EMF is zero at all speeds
- (D) Induced EMF depends only on temperature

Table 1: Experimental Observation Data

Observation Trial	Magnet Velocity (m/s)	Induced Current (mA)
Trial 1	0.5	12.4
Trial 2	1.0	24.8

4. Which rule is used to determine the direction of induced current?

- (A) Fleming's Right-Hand Rule (B) Fleming's Left-Hand Rule (C) Right-Hand Thumb Rule (D) Coulomb's Law

ANSWER KEY

1-B, 2-B, 3-A, 4-A